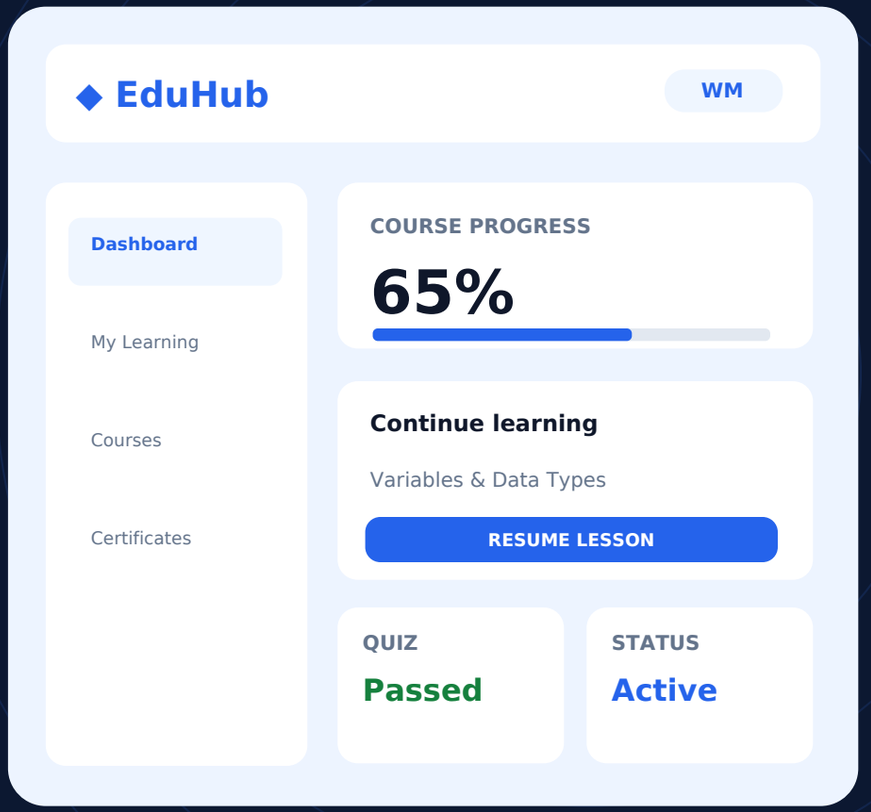
EDUHUB / DESIGN SPECIFICATION 02

EDUHUB

UI/UX Guidelines

Visual foundations, interaction conventions, accessibility, and responsive behavior for a role-based learning platform.



SYSTEM ANALYSIS & DESIGN

Companion to Wireframes & Mockups | 05 November 2026

Purpose, scope & sources

What is grounded in the current product design - and what requires team approval

Design governance

These guidelines standardize EduHub's interface across public pages and Student, Instructor and Admin workspaces. They are intended for designers, ASP.NET Core MVC developers, and reviewers.

FIGMA OBSERVED

Existing colors, Inter type, selected states, page regions, and 390 / 834 / 1440 px artboards.

PROJECT REQUIREMENT

Course discovery, content authoring, quizzes, payments, progress, certificates and role-based administration.

TEAM RECOMMENDATION

Accessible behavior, component naming, breakpoint mapping and consistent failure handling.

Authority model

F Existing EduHub Figma screens

P Uploaded 5-page project brief

S Proposed implementation convention

Adoption rule

Visual examples show expected presentation; they do not prove implementation. Treat recommendations as design decisions until the team reviews and accepts them.

Open EduHub Figma source

Target standard: WCAG 2.2 Level AA [W1]

Six operating principles

A consistent experience for every role, task and device

01

Clarity before density

State the task, show the next action, and explain unfamiliar platform terms.

LEARNER VALUE

02

Learning continuity

Preserve progress, resume lesson position and show the next achievable step.

LEARNER VALUE

03

One role, one workspace

Keep student, instructor and admin navigation separated by permission.

TRUST & CONTROL

04

Visible system status

Express loading, success, failure and pending outcomes in text and controls.

SYSTEM FEEDBACK

05

Safe commitments

Confirm destructive actions, preview purchases and prevent duplicate submissions.

RISK REDUCTION

06

Access for everyone

Support keyboard use, readable contrast, forms, captions and scalable content.

INCLUSION

Role-specific navigation & permissions

Navigation adapts to purpose; authorization remains a server-side responsibility

STUDENT

FIGMA

Navigation

- **Dashboard**
- Explore Courses
- My Learning
- My Certificates
- My Profile

Discover > enroll > watch > assess > certify

INSTRUCTOR

FIGMA

Navigation

- **Dashboard**
- My Courses
- Create Course
- Students
- Course Analytics

Draft > curriculum > preview > submit > monitor

ADMIN

FIGMA

Navigation

- **Dashboard**
- Course Approvals
- User Management
- Payments

Review > approve/reject > audit > manage

RULE: Hide forbidden actions in UI; enforce role/ownership checks in ASP.NET Identity + server-side authorization.

Color system & semantic meaning

Core hues sampled from representative public, student, instructor and admin Figma frames

PRIMARY
#2563EB
Actions / links

INK
#0F172A
Primary text

SECONDARY
#64748B
Supporting text

SURFACE
#FFFFFF
Cards / fields

CANVAS
#F8FAFC
App background

BORDER
#E2E8F0
Separators

SELECTED
#EFF6FF
Active navigation

HIGHLIGHT
#DBEAFE
Accent surface

SUCCESS
#15803D
Completed

SUCCESS BG
#F0FDF4
Success surface

WARNING
#92400E
Pending / warning

WARNING BG
#FFFBEB
Warning surface

DANGER
#B91C1C
Rejected / failure

DANGER BG
#FEF2F2
Error surface

Color is never the only signal

Pair every badge and progress bar with a readable label (e.g. Pending, 65% complete, Payment failed). Semantic use is proposed; shades are sampled from Figma.

Contrast evidence & corrections

Calculated from Figma hex values; normal text target >= 4.5:1, large text >= 3:1 [W1]

TEXT / BACKGROUND		RATIO	TEXT AA	SUGGESTED USE
Aa	White / primary blue	5.17:1	PASS	Buttons / links
Aa	Muted / white	4.76:1	PASS	Secondary labels
Aa	Muted / canvas	4.55:1	PASS	Body on canvas
Aa	Primary / selected	4.75:1	PASS	Active text
Aa	Primary / light blue	4.24:1	CAUTION	Normal text
Aa	Muted / selected	4.37:1	CAUTION	Normal text
Aa	White / ink	17.85:1	PASS	Inverse contrast

ACTION: Prefer dark ink on pale surfaces for normal-sized text; test every implemented state.

Typography, hierarchy & readability

Figma uses Inter in Regular, Medium, Semi Bold and Bold weights

TYPE RAMP

Display / hero	Learn more	40 px
Page title	Build your skills	28 px
Section heading	Continue learning	22 px
Card heading	Featured course	18 px
Body strong	Start with one lesson.	14 px
Body / controls	View course details	13 px
Supporting copy	Your course starts here.	12 px
Micro-label	Updated just now	11 px

Typography usage

Recommended mapping of observed Figma type: headings 28-40 px; sections 18-22 px; most controls 13-14 px. The source file contains additional values across layouts.

FONT FAMILY

Inter / system-ui / sans-serif

Readability standards

Line length

Use approximately 45-75 characters per line.

Hierarchy

One page H1; meaningful H2/H3 structure.

Scaling

Allow 200% zoom without lost actions.

Microcopy

Short, specific and locally understandable.

Spacing, corners & elevation

Observed roundness is retained; spacing scale below is a team implementation recommendation

Spacing system

RECOMMENDED SCALE

XS	4px	Icon gap
S	8px	Inline gap
M	12px	Control inset
L	16px	Card inset
XL	24px	Section gap
2XL	32px	Page gutter
3XL	48px	Major section

Corner radius

4px	8px	12px	100px
Small control	Card / input	Container	Pill

Elevation & surfaces

Use subtle depth

Prefer borders and surface contrast to large shadows. Use stronger elevation only for dialogs and overlays.

Layouts adapt, priorities remain

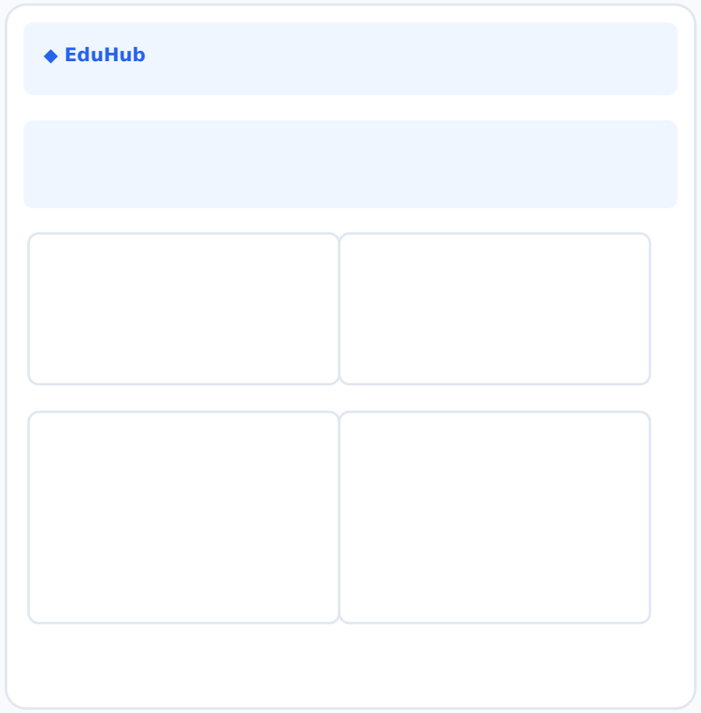
Figma artboards: 390 mobile, 834 tablet, 1440 desktop; Bootstrap breakpoints are separate [B1]

390 px FIGMA



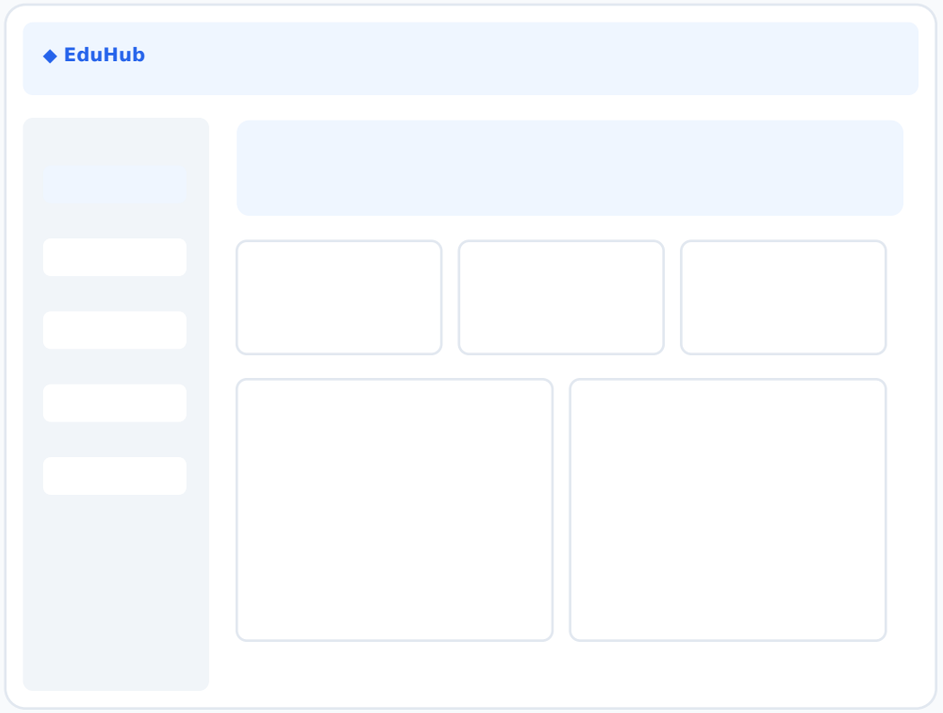
MOBILE

834 px FIGMA



TABLET

1440 px FIGMA



DESKTOP

BREAKPOINT RULE Bootstrap: xs <576, sm >=576, md >=768, lg >=992, xl >=1200, xxl >=1400 px. Design content-first, not device-name-first.

Navigation & wayfinding

Consistent location cues, meaningful labels and predictable back paths

Patterns to use

◆ EduHub

STUDENT WORKSPACE

- Dashboard
- Explore Courses
- My Learning
- My Certificates
- My Profile

Home / My Learning / Complete C# Programming

Variables & Data Types

CURRENT PAGE: Course Learning

A clear breadcrumb supports navigation and recovery.

On mobile, keep core destinations visible or reachable with a labeled menu; do not rely on icons alone. On desktop, preserve an active sidebar item. Always offer a safe back route.

Behavior contracts

Global search

Search field has accessible label, submit action and no-results state.

Active page

Current nav item visually highlighted and marked aria-current.

Breadcrumb

Only ancestors are links; final crumb identifies current page.

Role boundary

Inaccessible routes yield a respectful access-denied screen.

Mobile menu

Labeled button exposes state via aria-expanded and closes by Escape.

Buttons, links & action hierarchy

A single dominant action per decision region; states must remain visually and programmatically clear

Action specimens

FIGMA-ALIGNED

Continue Learning	View Course	Delete Course
PRIMARY	SECONDARY	DESTRUCTIVE
Saving…	Save as Draft	Submit for Review
DISABLED / BUSY	SAFE ALTERNATIVE	FOCUS EXAMPLE

Use verbs naming the effect: 'Enroll for Free', 'Complete Purchase', 'Mark as Completed', 'Submit for Review'. Avoid vague 'OK' labels. Disable repeated payment submission during processing.

Component specification

Height

44-48 px observed often; 44 px proposed touch target.

Focus

Clear, high-contrast ring, not just a color shift.

Processing

Spinner + unchanged action context; avoid duplicate requests.

Destructive

Confirm deletion/refund where not easily reversible.

Disabled

Never use low contrast to hide required explanation.

Links

Underlined or identifiable without color alone.

Forms, validation & upload

Required fields and corrective feedback must be accessible and actionable [W2][W3]

Form specimen: course creation

Course Title *

Complete C# Programming

Keep the title specific and easy to find.

Course Price

-5.00

Enter a price of zero or higher.

Course Thumbnail

Drop JPG or PNG here

Max 5 MB • 1280 × 720 suggested

Always show labels above inputs. Link inline errors to the field and include a top-level summary for multi-error submissions. Validate again on the server.

Implementation criteria

Labels

Use <label for='id'>. Placeholder is not a label.

Required

Mark visually and programmatically with required.

Errors

Explain what failed and how to correct it.

Descriptions

Bind help/error text via aria-describedby.

Input types

Use email, number, URL, file as appropriate.

Uploads

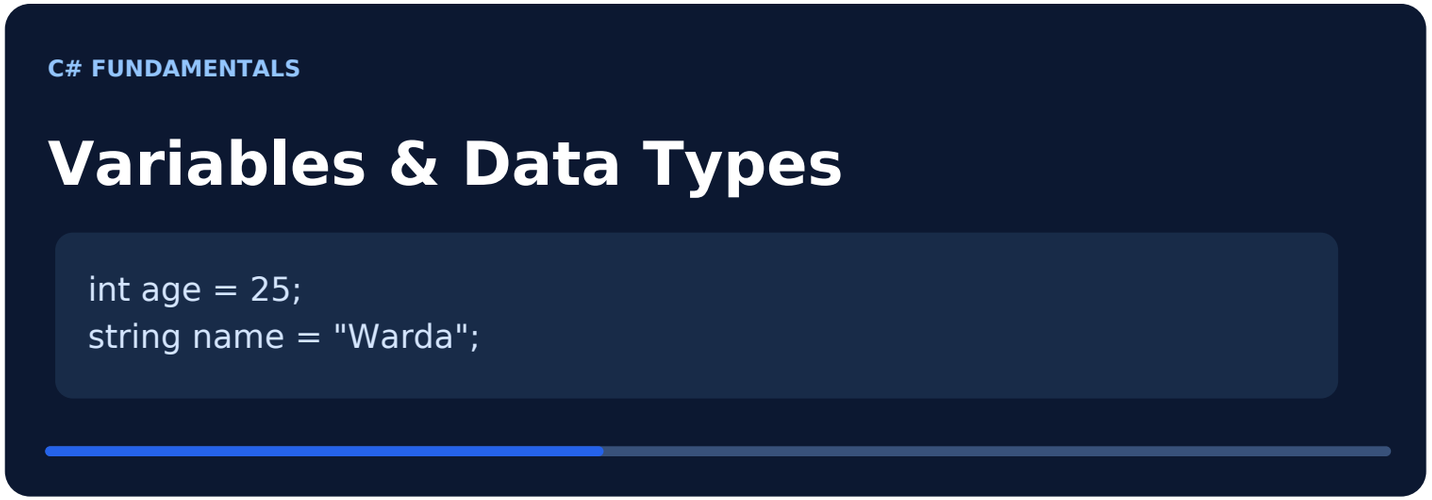
Show type/size/processing status; support retry.

Figma: instructor course form [F]; WAI [W2][W3]

Video, learning progress & notes

Make learning status visible and keep playback usable with keyboard and captions

Learning player example



Learning control rules

Player

Use native accessible controls or verified Video.js keyboard support.

Captions

Provide accurate captions when prerecorded speech matters [W5].

Resume

Save watch position; disclose the resume action.

Completion

Display numeric and textual status, not just bar color.

Notes / bookmarks

Persist against the lesson and indicate save success.

Assessments

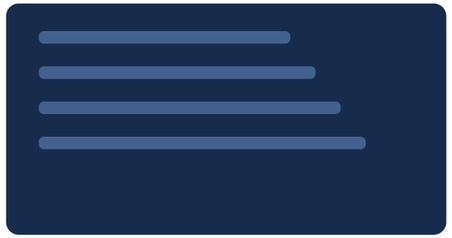
Show eligibility, attempts and the passing threshold.

Figma: course learning [F]; captions [W5]

Discovery cards, filters & data tables

Course selection should be easy to scan; admin data should be easy to act on

Course card specimen



PROGRAMMING

Complete C# Programming

Ahmed Mohamed

4.8 (1.2k) • Beginner

$49.00

View Course

Filter structure

- Search title
- Category
- Level
- Price
- Rating

Filters must show selected values, support reset and expose an informative empty result.

Data table pattern

COURSE	STATUS	ACTION
SQL Server Essentials	Pending	Review
Complete C# Programming	Approved	View
REST APIs with .NET	Rejected	Review

On mobile, prefer labeled stacked records over a compressed seven-column table. Preserve sorting, filtering, pagination and explicit status labels.

Assessment, outcomes & certificates

Give clear eligibility, review, result feedback and verifiable proof of completion

Quiz interaction contract

10 questions • Passing score 70% • Review before submit

What keyword defines a class in C#?

- ○ function
- ◉ **class**
- ○ object
- ○ method

Previous Question | **Next Question**

Flow and outcome behavior

01 Access
Quiz remains locked until course completion policy is met.

02 Answer
Selected option persists when moving between questions.

03 Review
Question navigation marks answered / unanswered clearly.

04 Submit
Show confirmation, then scored result and explanations.

05 Certify
Only eligible completions issue downloadable certificates.

06 Verify
Use a unique code and public verification page [P].

Course authoring & publishing

Separate drafting from public release; make upload progress and review decisions obvious

Course lifecycle

MAIN HAPPY PATH - implementation concept derived from brief + source screen states



Key instructor rules

Draft persistence	Save explicitly, show 'Saved' timestamp and warn on unsaved navigation.
Curriculum ordering	Drag handles need keyboard-based reorder and move alternatives.
Media upload	Validate file type and size, show progress, processing and retry.
Review handling	Submitted content must indicate pending, approval, rejection and feedback.

Course builder controls

≡ **Section: Introduction to C#**

≡ Lesson: Welcome to the Course

Save Draft

Before publish: a clear review summary and status.

Reviews, governance & sensitive actions

Administrative interfaces must prioritize accuracy, authorization and auditability

Approval decision panel

COURSE REVIEW / SQL Server Essentials

Submitted by

Mohamed Hassan

Current status

Pending

Review checks

- [ ] Course title, category and description
- [ ] Lessons, video quality and completeness
- [ ] Pricing, visibility and policy compliance

Approve Course | **Reject with Reason**

Admin decision policy

Explicit effect

Labels state approve/reject and impact on visibility.

Reason capture

Rejection should require an actionable reason.

Confirmation

Use confirm step for refunds / suspensions / deletions.

Authorization

Only privileged roles access actions; check server-side.

Audit trail

Record reviewer, decision, timestamp and reason.

Feedback

Show success/error state and update counts and queues.

Checkout trust, payment status & retry

The Figma checkout is labeled demo-only; production payment behavior needs explicit approval

Checkout layout - behavior

DEMO CHECKOUT - no real card information

Payment Information

Payment method

Cardholder name

Card details

Order Summary

Course price **$49.00**

Total **$49.00**

Complete Purchase

Before redirecting or confirming, display the final total, any discounts and a clear indication of real vs demo payment.

Status & security requirements

Processing

Show spinner; don't create duplicate orders on retries.

Success

Confirm enrollment only after server-side payment verification.

Failure

Preserve cart; give safe retry or alternative next step.

Pending

Never imply enrollment is confirmed while status is unknown.

Production

Use Stripe's secure hosted / embedded payment flow.

Protection

Never log/store raw card numbers or CVV in EduHub.

System states & feedback language

Every asynchronous or blocked interaction has an explicit, non-color-only state

✓ Payment complete

SUCCESS

Your enrollment is confirmed.

Next: open course

! Awaiting approval

WARNING

Course remains hidden while reviewed.

Next: view submission

✕ Could not save

ERROR

Your changes are still on this page. Try again.

Next: retry save

i Processing

INFO

Checking payment status; do not close this page.

Next: wait / refresh

– No courses yet

EMPTY

Create a course to get started.

Next: create course

! Access denied

FORBIDDEN

Return to your permitted dashboard.

Next: return to dashboard

WCAG 2.2 AA: minimum review gates

These are testable acceptance criteria, not a claim that the current design already conforms [W1]

SC	TEST AREA	ACCEPTANCE CHECK	
1.4.3	**Text contrast**	Normal >=4.5:1; large >=3:1.	○
1.4.11	**Non-text contrast**	Important borders and state cues >=3:1.	○
2.4.7	**Focus visible**	Tab focus clearly visible on every actionable element.	○
2.5.8	**Target size**	24×24 CSS px AA minimum, with exceptions.	○
1.4.10	**Reflow**	No loss / overlap at 320 CSS px, except allowed cases.	○
1.4.4	**Resize text**	At 200% text resize, features remain operable.	○
3.3.1	**Error identification**	Errors identified in text and linked to fields.	○
3.3.2	**Labels / instructions**	Every input has name and required guidance.	○
1.2.2	**Captions**	Captions for prerecorded synchronized media with audio.	○
2.1.1	**Keyboard**	Core tasks operable without a mouse.	○

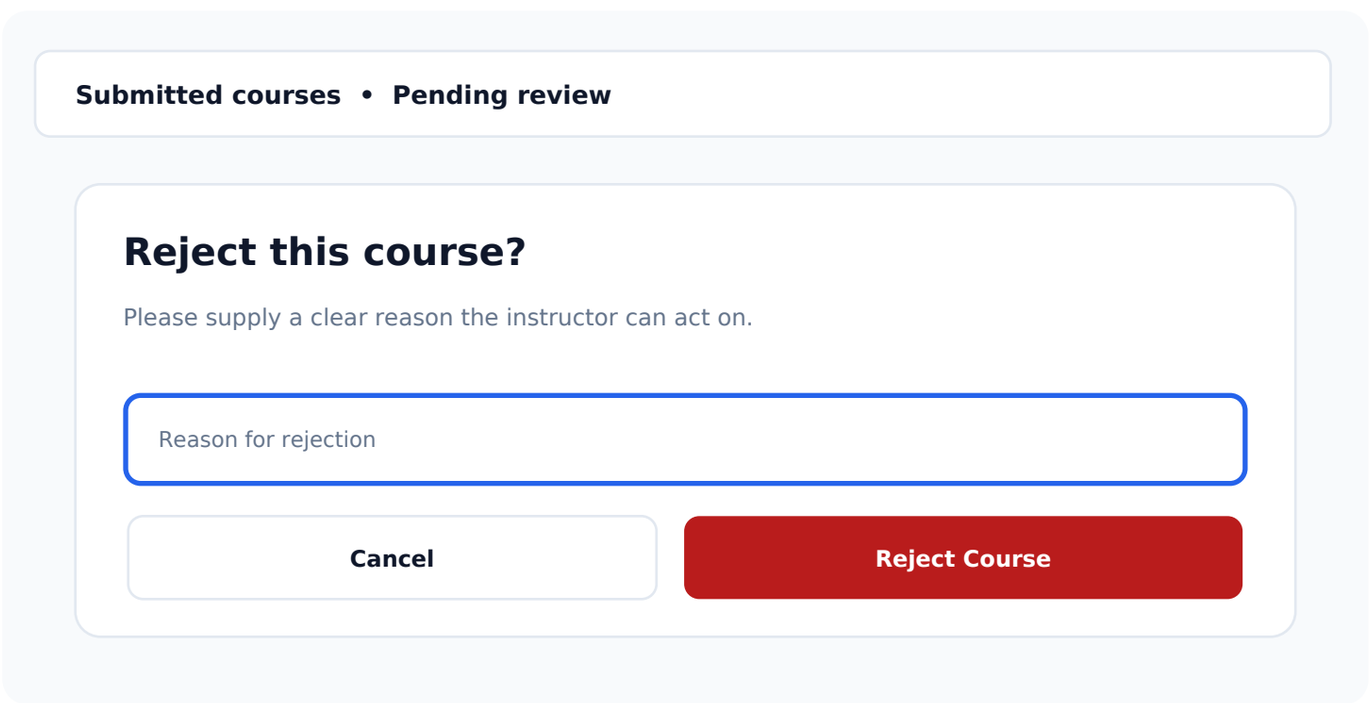
Recommended application target: 44×44 px for frequent touch controls; this is a team convention, not WCAG AA minimum.

Source: WCAG 2.2 [W1] / WAI [W2][W3]

Keyboard, focus & modal behavior

Use native HTML controls first; custom patterns must follow WAI-ARIA behavior [W4]

Keyboard scenario: course approval



FOCUS EXAMPLE: 2 px high-visibility outline on active field.

Keyboard behavior checklist

Tab / Shift+Tab

Move through actionable controls in a logical reading order.

Enter / Space

Activate button controls; space toggles checkboxes.

Escape

Close modal/menu when allowed; retain recoverability.

Modal focus

Move focus inside, trap focus while open, restore on close.

Reorder lists

Provide Move up / Move down alternative to drag & drop.

Status changes

Announce async results using appropriate live regions.

WAI Dialog APG [W4]; Figma admin states [F]

UX writing, messages & terminology

Use direct and consistent language that helps users recover and continue

Before / after writing examples

Vague CTA	Click here	Continue Learning
Poor error	Invalid data	Enter a price of zero or higher.
No context	Failed	Payment failed. Your card wasn't charged.
Unclear status	Done	Course submitted for admin review.
Quiz feedback	Wrong	Not quite. The correct answer is class.

Voice, terminology & formatting

Tone

Clear, professional, encouraging; avoid blame or hype.

Course

Use one noun for course; section / lesson consistently.

Student actions

'Enroll for Free', 'Resume Lesson', 'Take Final Quiz'.

Instructor actions

'Save Draft', 'Submit for Review', 'Publish' only when valid.

Admin actions

'Approve Course', 'Reject with Reason', 'Issue Refund'.

Amounts / dates

Consistent currency symbols, locale, and time zone.

Localization

Keep strings in resources; plan English/Arabic and RTL if needed.

Loading, feedback timing & resilience

Communicate change without making the user depend on motion or color

State progression

IDLE
Await action

LOADING
Disable duplicate action

SUCCESS
Show confirmation

ERROR
Retry, preserve input

Motion recommendations [S]

Use brief transitions (about 120-220 ms) for nonessential state changes. Respect prefers-reduced-motion. Don't animate essential status indefinitely without accessible text or exit.

Asynchronous completion should update the status label and focus/announcement.

Operational UX

Uploads

Show bytes/progress, processing step, successful URL or retry.

Video save

Show position saved without blocking playback.

Quiz submit

Prevent double submit; return score or actionable error.

Payment

Use order status on reload; no optimistic enrollment.

Notifications

Distinguish unread/read; allow retry if refresh fails.

Search

Show empty result, active filter chips and clear-all route.

Design tokens & Bootstrap mapping

Suggested starting contract for ASP.NET Core MVC Razor views, Bootstrap 5 and shared CSS

CSS variable proposal

```
:root {
  --eh-primary: #2563EB;
  --eh-ink: #0F172A;
  --eh-muted: #64748B;
  --eh-canvas: #F8FAFC;
  --eh-surface: #FFFFFF;
  --eh-border: #E2E8F0;
  --eh-selected: #EFF6FF;
  --eh-success: #15803D;
  --eh-warning: #92400E;
  --eh-danger: #B91C1C;
  --eh-radius-sm: 4px;
  --eh-radius-md: 8px;
  --eh-radius-lg: 12px;
  --eh-control-h: 48px;
  --eh-font: Inter, system-ui, sans-serif;
}
```

Delivery conventions

Shared layout

Use role-aware Razor _Layout and reusable partials.

Bootstrap

Map grid classes and utility tokens, keep overrides scoped.

Components

One source for button, badge, field, alert and dialog.

Form errors

ASP.NET validation summary + field-level errors.

Names

CSS: .eh-card, .eh-status; Razor partials: _CourseCard.

Responsive QA

Test 390 / 834 / 1440 and between sizes.

Status source

Render labels from authoritative server state.

Token values observed; CSS contract recommended [S]

Core component inventory & ownership

Reusable UI elements, expected states and relevant source frame

COMPONENT	STATE CONTRACT	OWNER	FIGMA REF
App shell / sidebar	active, collapsed, unauthorized	Shared layout	**9:7335**
Course card	free, paid, enrolled, unavailable	Catalog partial	**22:38019**
Course progress	0..100%, completed, sync error	Student partial	**22:39539**
Quiz question	unanswered, selected, submitted	Assessment view	**22:40004**
Certificate card	locked, issued, verified	Student view	**22:40308**
Course form	pristine, dirty, invalid, saving	Instructor view	**18:56939**
Media uploader	empty, uploading, processing, error	Instructor view	**18:58461**
Approval row	pending, approved, rejected	Admin view	**26:84906**
Checkout summary	pricing, submitting, paid, failed	Payment view	**22:39207**
Status alert	info, success, warning, error	Shared partial	**70:43449**

Traceability is design-level. Technical identifiers/routes remain subject to the system architecture documentation.

UI acceptance & validation checklist

Suggested minimum exit gate for the System Analysis & Design submission and implementation review

- [ ] **Visual fidelity**
  Component colors, typography, spacing match approved Figma variants.
- [ ] **Viewport coverage**
  390, 834, 1440 px and intermediate widths show no breakage.
- [ ] **Role safety**
  Student / Instructor / Admin display only permitted actions.
- [ ] **Keyboard**
  Every core user flow works with Tab, Enter, Space and Escape.
- [ ] **Focus**
  Visible indicator; modal focus enters, stays, and restores correctly.
- [ ] **Text contrast**
  Text and meaningful UI cues meet WCAG 2.2 AA thresholds.
- [ ] **Forms**
  Labels, hints, required fields and inline/server errors available.
- [ ] **Empty / loading**
  All data regions have no-data, loading, failure and retry states.
- [ ] **Video**
  Captions and player keyboard controls verified where applicable.
- [ ] **Checkout**
  Total reviewed, duplicate submit prevented, status server-verified.
- [ ] **Quiz / certificate**
  Pass/fail, eligible/locked and verification paths tested.
- [ ] **Responsive nav**
  Mobile navigation and content reflow remain navigable.

Design questions for team sign-off

These are design / product decisions, not claims that Figma is incorrect

ID	DECISION	RECOMMENDED DISCUSSION	RISK
D-01	**Public Admin registration**	Figma signup includes Admin role choice. Decide whether to restrict Admin creation to invited/authorized accounts.	High
D-02	**Demo vs. live checkout**	Current visual includes demo-only card-entry warning. Confirm Stripe production experience and status verification.	High
D-03	**Certificates**	Brief calls for public verification code/page. Agree on URL, no-data, invalid-code and privacy behavior.	High
D-04	**Q&A and announcements**	Brief mentions instructor Q&A and announcements. Confirm detailed UI coverage and states.	Medium
D-05	**Accessibility completion**	Screens exist, but code-level keyboard, captions, focus, contrast and zoom need actual testing.	High
D-06	**Internationalization**	Decide scope for Arabic/RTL, content localization and currency/timezone formatting.	Medium

APPROVAL CRITERION: Record owner, decision, effective date and Figma / backlog update for every agreed change.

Evidence register & linked standards

Verifiable primary sources for this guideline document

F	**EduHub Figma design**	Source screens, palette, typography, layouts, role workspaces.
P	**EduHub Project Details PDF**	User-provided 5-page graduation project brief (local attachment).
W1	**W3C - WCAG 2.2**	Accessibility success criteria and levels A/AA.
W2	**W3C WAI - Labeling Controls**	Form labels and programmatic association.
W3	**W3C WAI - Validating Input**	Accessible field validation and prevention.
W4	**W3C APG - Modal Dialog Pattern**	Focus trapping, Escape and restoration.
W5	**W3C WAI - Captions/Subtitles**	Captions for prerecorded video with meaningful audio.
B1	**Bootstrap 5 - Responsive breakpoints**	Official six-tier responsive viewport thresholds.

Internal source and recommendations are intentionally distinguished throughout the guide.